192425181

R.Poornima siva priya

1. Execute and display the results for the following commands in Kali Linux Operating

Systems:

(a) rm command

(b) users command

(c) tree command

Aim:
To learn and perform the rm, users, and tree commands in Kali Linux and observe their outputs.

```
labex:/tmp/ $ cd /tmp
labex:/tmp/ $ touch testfile.txt
labex:/tmp/ $ ls testfile.txt
testfile.txt
labex:/tmp/ $ rm testfile.txt
labex:/tmp/ $ ls testfile.txt
ls: cannot access 'testfile.txt': No such file or directory
```

```
labex:/tmp/ $ users
labex:/tmp/ $
```

```
labex:/tmp/ $ cd /tmp
labex:/tmp/ $ mkdir -p mytree/folder1
labex:/tmp/ $ mkdir -p mytree/folder2
labex:/tmp/ $ touch mytree/file1.txt
labex:/tmp/ $ touch mytree/folder1/file2.txt
labex:/tmp/ $ touch mytree/folder2/file3.txt
labex:/tmp/ $ tree mytree
mytree
|-- file1.txt
|-- folder1
|   `-- file2.txt
`-- folder2
    `-- file3.txt

2 directories, 3 files
```

2. Execute and display the results for the following commands in Kali Linux Operating

Systems:

(a) less command

(b) more command

(c) vi command

```
labex:/tmp/ $ less sample.txt
labex:/tmp/ $ more sample.txt
Line 1: Kali Linux
Line 2: Operating System
Line 3: Linux Commands
Line 4: File Management
Line 5: Terminal
Line 6: Lab Exercise
Line 7: Less Command
Line 8: More Command
Line 9: Vi Editor
Line 10: End of File
labex:/tmp/ $ vi vi_demo.txt
labex:/tmp/ $ cat vi_demo.txt
Kali Linux Operating System
This is a vi editor demonstration.
Linux commands are useful.
```

3. Execute and display the results for the following commands in Kali Linux Operating Systems:

(a) Tree

(b) Head

(c) Tail

```
labex:/tmp/ $ cd /tmp
labex:/tmp/ $ mkdir -p tree_demo/folder1
labex:/tmp/ $ mkdir -p tree_demo/folder2
labex:/tmp/ $ touch tree_demo/file1.txt
labex:/tmp/ $ touch tree_demo/folder1/file2.txt
labex:/tmp/ $ touch tree_demo/folder2/file3.txt
labex:/tmp/ $ tree tree_demo
tree_demo
|-- file1.txt
|-- folder1
|   `-- file2.txt
`-- folder2
    `-- file3.txt

2 directories, 3 files
```

```
labex:/tmp/ $ head -5 commands.txt
Line 1: Kali Linux
Line 2: Operating System
Line 3: Linux Commands
Line 4: File Management
Line 5: Terminal
```

```
labex:/tmp/ $ tail -5 commands.txt
Line 6: Lab Exercise
Line 7: Head Command
Line 8: Tail Command
Line 9: Tree Command
Line 10: End of File
labex:/tmp/ $
```

4. Execute and display the results for the following commands in Kali Linux Operating

Systems:

i) Ls

ii) Cat

iii) Mkdir

Aim:

To execute and display the results of ls, cat, and mkdir commands in Kali Linux.

```
labex:/tmp/ $ cd /tmp
labex:/tmp/ $
labex:/tmp/ $ ls
commands.txt        mytree            testfile.txt  vi_d
hsperfdata_root     sample.txt        tmux-5000
monitor.log         session           tree_demo
monitor_session.sh  ssh-XXXXXXstQpit  vDgYONsZ
```

```
labex:/tmp/ $ printf "Welcome to Kali Linux\nThis is a Linux practical.\nUs
ing the cat command.\n" > cat_demo.txt
labex:/tmp/ $ cat cat_demo.txt
Welcome to Kali Linux
This is a Linux practical.
Using the cat command.
```

```
labex:/tmp/ $ mkdir my_directory
labex:/tmp/ $ ls
cat_demo.txt     monitor_session.sh  session           tree_demo
commands.txt     my_directory        ssh-XXXXXXstQpit  vDgYONsZ
hsperfdata_root  mytree              testfile.txt      vi_demo.txt
monitor.log      sample.txt          tmux-5000
labex:/tmp/ $
```

5. Execute and display the results for the following commands in Kali Linux Operating

Systems:

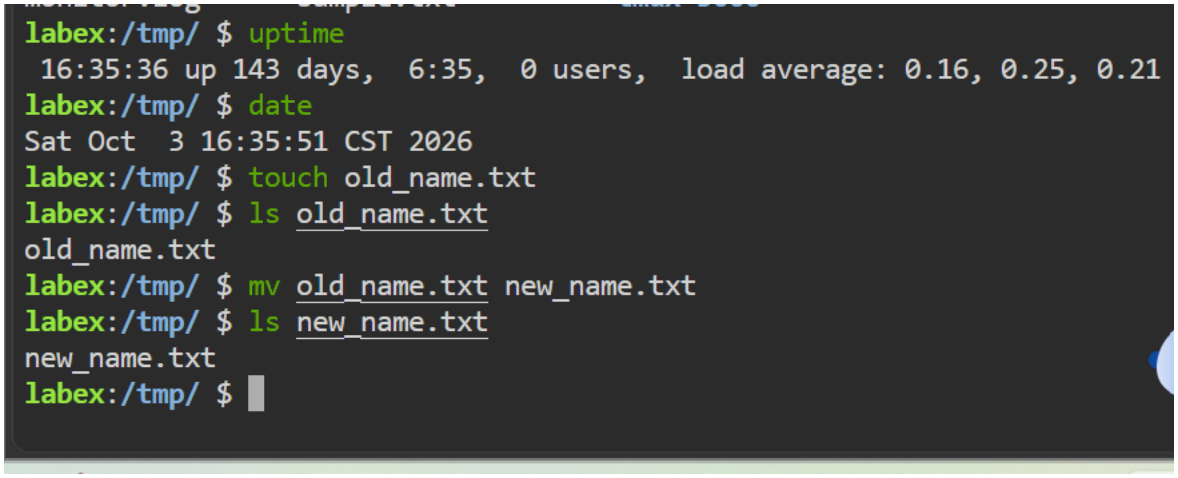
i) Uptime

ii) Date

iii) Mv

Aim:

To execute and display the results of uptime, date, and mv commands in Kali Linux.



6. Execute and display the results for the following commands in Kali Linux Operating

Systems:

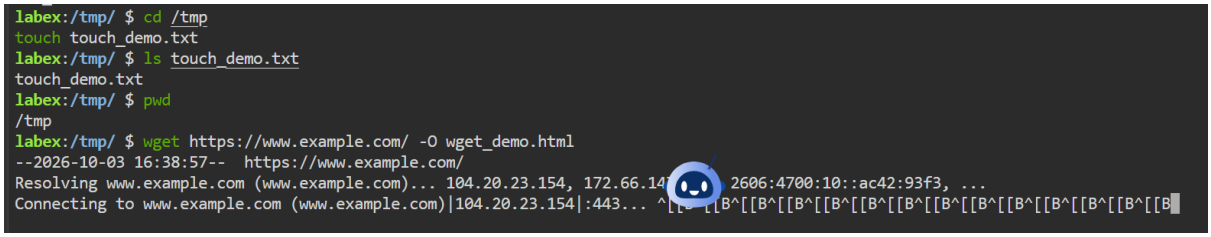
i) Touch

ii) Pwd

iii) Wget

Aim:

To execute and display the results of touch, pwd, and wget commands in Kali Linux.



7. Execute and display the results for the following commands in Kali Linux Operating

Systems:

Cd

Cp

ls

```
labex:project/ $ cd /tmp
labex:/tmp/ $ pwd
/tmp
labex:/tmp/ $ echo "This is a copy command demonstration." > original.txt
labex:/tmp/ $ cp original.txt copy.txt
labex:/tmp/ $ ls
LxCLHGHL  hsperfdata_root  monitor_session.sh  session           tmux-5000
copy.txt  monitor.log      original.txt        ssh-XXXXXXQpYhui
labex:/tmp/ $ cat copy.txt
This is a copy command demonstration.
labex:/tmp/ $ ls
LxCLHGHL  hsperfdata_root  monitor_session.sh  session           tmux-5000
copy.txt  monitor.log      original.txt        ssh-XXXXXXQpYhui
labex:/tmp/ $
```

8. Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems:

i) detect the version of services running

ii) aggressive scan

iii) detect operating system of the target

```
22/tcp   open  ssh         OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Linux; protocol 2.0)
3001/tcp open  ssl/nessus?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint
cgi?new-service :
SF-Port3001-TCP:V=7.80%T=SSL%I=7%D=10/3%Time=6AC0C58B%P=x86_64-pc-linux-gn
SF:u%r(GetRequest,36E7,"HTTP/1\.1\x20200\x20OK\r\nServer:\x20KasmVNC/4\.0\
SF:r\nConnection:\x20close\r\nContent-type:\x20text/html\r\nContent-length
SF::\x2013863\r\nCross-Origin-Embedder-Policy:\x20require-corp\r\nCross-Or
SF:igin-Opener-Policy:\x20same-origin\r\n\r\n<!doctype\x20html><html\x20la
SF:ng=en\x20class=noVNC_loading><title>KasmVNC</title><meta\x20charset=utf
SF:-8><meta\x20name=viewport\x20content=\"width=device-width,initial-scale
SF:=1,maximum-scale=1,user-scalable=no\"><meta\x20name=apple-mobile-web-ap
SF:p-capable\x20content=yes><meta\x20name=apple-mobile-web-app-status-bar-
SF:style\x20content=black-translucent><script\x20src=vendor/interact\.min\
SF:.js></script><link\x20href=dist/style\.bundle\.css\x20rel=stylesheet><d
SF:iv\x20id=noVNC_fallback_error\x20class=noVNC_center><div><div>noVNC\x20
SF:encountered\x20an\x20error:</div><br><div\x20id=noVNC_fallback_errormsg
SF:></div></div></div><div\x20id=noVNC_connection_stats>Loading\x20statist
SF:ics\.\.\.</div><div\x20id=noVNC_control_bar_anchor\x20class=noVNC_vcent
SF:er\x20style=\"display:none\"><div\x20id=noVNC_control_b");
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 152.55 seconds
```

```
cgi?new-service :
SF-Port3001-TCP:V=7.80%T=SSL%I=7%D=10/3%Time=6AC0C6F7%P=x86_64-pc-linux-gn
SF:u%r(GetRequest,36E7,"HTTP/1\.1\x20200\x20OK\r\nServer:\x20KasmVNC/4\.0\
SF:r\nConnection:\x20close\r\nContent-type:\x20text/html\r\nContent-length
SF::\x2013863\r\nCross-Origin-Embedder-Policy:\x20require-corp\r\nCross-Or
SF:igin-Opener-Policy:\x20same-origin\r\n\r\n<!doctype\x20html><html\x20la
SF:ng=en\x20class=noVNC_loading><title>KasmVNC</title><meta\x20charset=utf
SF:-8><meta\x20name=viewport\x20content=\"width=device-width,initial-scale
SF:=1,maximum-scale=1,user-scalable=no\"><meta\x20name=apple-mobile-web-ap
SF:p-capable\x20content=yes><meta\x20name=apple-mobile-web-app-status-bar-
SF:style\x20content=black-translucent><script\x20src=vendor/interact\.min\
SF:.js></script><link\x20href=dist/style\.bundle\.css\x20rel=stylesheet><d
SF:iv\x20id=noVNC_fallback_error\x20class=noVNC_center><div><div>noVNC\x20
SF:encountered\x20an\x20error:</div><br><div\x20id=noVNC_fallback_errormsg
SF:></div></div></div><div\x20id=noVNC_connection_stats>Loading\x20statist
SF:ics\.\.\.</div><div\x20id=noVNC_control_bar_anchor\x20class=noVNC_vcent
SF:er\x20style=\"display:none\"><div\x20id=noVNC_control_b");
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submi
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
421 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@6ac124d8d217b2dc24ef440c:~# nmap --version
Nmap version 7.80 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.2 nmap-libssh2-1.8.2 libz-1.2.11 libpcre-8.39 libpcap-1.10.1 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
root@6ac124d8d217b2dc24ef440c:~# nmap -O 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-04 00:06 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000012s latency).
Not shown: 998 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
3001/tcp open  nessus
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.47 seconds
root@6ac124d8d217b2dc24ef440c:~#
```

9. Execute and display the results for the following Nmap Timing and performance commands in Kali Linux operating Systems:

(i) polite IDS evasion

(ii) normal IDS evasion

```
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.47 seconds
root@6ac124d8d217b2dc24ef440c:~# nmap -T2 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-04 00:11 CST
Stats: 0:02:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 37.10% done; ETC: 00:18 (0:04:13 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 37.40% done; ETC: 00:18 (0:04:11 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 37.50% done; ETC: 00:18 (0:04:12 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 37.60% done; ETC: 00:18 (0:04:11 remaining)
Stats: 0:02:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.10% done; ETC: 00:18 (0:04:09 remaining)
Stats: 0:02:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.20% done; ETC: 00:18 (0:04:08 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000031s latency).
Not shown: 998 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
3001/tcp open  nessus

Nmap done: 1 IP address (1 host up) scanned in 400.42 seconds
```

```
root@6ac124d8d217b2dc24ef440c:~# nmap -T3 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-04 00:31 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 998 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
3001/tcp open  nessus

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

10. To execute the following commands in Kali Linux Operating Systems:

Free command

(a) Sort command

(b) History command

11. Execute and display the results for the following Nmap Timing and performance commands in Kali Linux operating Systems:

(i) aggressive speed scan

(ii) insane speed scan

```
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
root@6ac124d8d217b2dc24ef440c:~# free -h
               total        used        free      shared  buff/cache   available
Mem:            14Gi       4.6Gi       5.0Gi        75Mi       5.4Gi        10Gi
Swap:             0B          0B          0B
root@6ac124d8d217b2dc24ef440c:~#
```

```
root@6ac124d8d217b2dc24ef440c:~# sort fruits.txt
apple
banana
grape
mango
orange
root@6ac124d8d217b2dc24ef440c:~#
```

```
root@6ac124d8d217b2dc24ef440c:~# history
    1  apt update
    2  apt install -y nmap
    3  apt update
    4  nmap --version
    5  nmap -O 127.0.0.1
    6  nmap -T2 127.0.0.1
    7  nmap -T2 127.0.0.10;276;0c
    8  nmap -T3 127.0.0.1
    9  free -h
   10  printf "banana\napple\norange\nmango\ngrape\n" > fruits.txt
   11  sort fruits.txt
   12  history
root@6ac124d8d217b2dc24ef440c:~#
```

11. Execute and display the results for the following Nmap Timing and performance commands in Kali Linux operating Systems:

(i) aggressive speed scan

(ii) insane speed scan

```
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
root@6ac124d8d217b2dc24ef440c:~# nmap -T5 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-04 00:46 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 998 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
3001/tcp open  nessus

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```